

CO₂ - ASSESSMENT

TOOL-1

Analytical Problem Solving

Name : K. Divya

Reg no : 192511316

Course : Data Structure

Code : CSA 0307

1. Double Linked List

Given elements : 150, 250, 350, 450, 550

After Insertion :

NULL

↓

[150] ⇌ [250] ⇌ [350] ⇌ [450] ⇌
[550]

↓

NULL

Each node contains:

[Previous | Data | Next]

Delete 150

150 is the first node, so after deleting it head moves to 250

NULL

↓

[250] ⇌ [350] ⇌ [450] ⇌ [550] → NULL

Delete 250.

250 is now the first node, so head moves to 350.

NULL

↓

[350] ⇌ [450] ⇌ [550] → NULL

Final Linked structure

$\text{NULL} \leftrightarrow [350] \leftrightarrow [450] \leftrightarrow [550] \rightarrow \text{NULL}$

Here:

$\Rightarrow$ 150 is the first node, so its prev = NULL

$\Rightarrow$ 550 is the last node, so its next = NULL

$\Rightarrow$ Every middle node has both prev and next links.

Thus, after inserting 150, 250, 350, 450, 550 and deleting 150 and 250, the remaining doubly linked list is:

$350 \rightleftharpoons 450 \rightleftharpoons 550$.

2. Binary Search

Given

Mr. XYZ wants to search for the PNR

number: 4056786

The ticket numbers are

4056779, 4056780, 4056781, 4056782,

4056783, 4056784, 4056785, 4056786, 4056787,

4056788, 4056789.

Since the ticket numbers are already arranged in ascending order, we can use Binary Search.

Binary Search Principle.

Binary Search repeatedly divides the search interval into two halves.

Formula:

$$\text{mid} = (\text{low} + \text{high}) / 2$$

If:

search value > Mid value

we search the right half.

If:

search value < Mid value.

we search the left half.

Step 1

$$\text{Low} = 1$$

$$\text{High} = 11$$

$$\begin{aligned}\text{Mid} &= (1 + 11) / 2 \\ &= 6\end{aligned}$$

Position 6 contains:

4056784.

Compare:

$$4056786 > 4056784.$$

Therefore, search the right half.

New range:

$$\text{Low} = 7$$

$$\text{High} = 11$$

Step 2 :

$$\begin{aligned}\text{Mid} &= (7+11)/2 \\ &= 9\end{aligned}$$

Position 9 contains:

4056787.

Compare:

$$4056786 < 4056787$$

Therefore, search the left half.

New range:

$$\text{Low} = 7$$

$$\text{High} = 8$$

Step 3:

$$\text{Mid} = (7+8)/2 = 7$$

Position 7 contains: 4056785

Compare $4056786 > 4056785$.

Therefore, search the right side.

New range: Low = 8, High = 8.

Step 4:

$$\text{Mid} = (8+8)/2 = 8$$

Position 8 contains 4056786

Now: $4056786 = 4056786$.

Therefore, the required PNR is found successfully.